

SECSR2043 OPERATING SYSTEMS
[20 Marks]

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 Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *.; cp ../fruits/*. *.;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

1. Create the woo.sh file

```
woo@secr2043:~$ nano woo.sh
```

2. Insert the code of the script provided

```
GNU nano 7.2 woo.sh *
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.*.; cp ../fruits/*.*.;
cp ../*.jar .
```

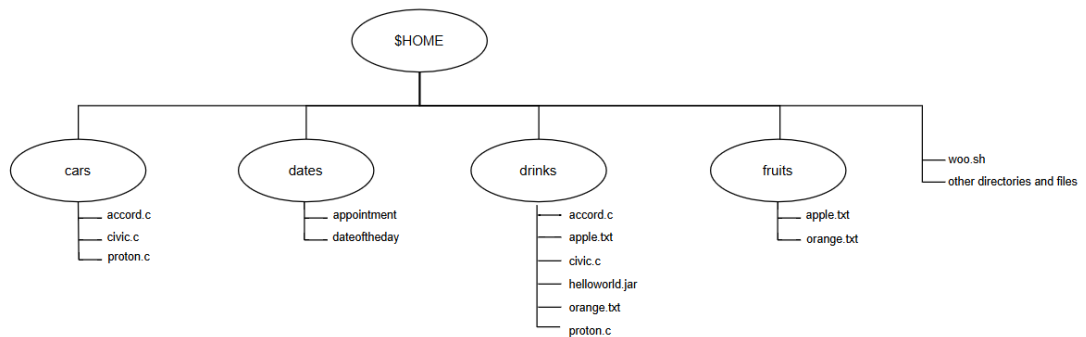
3. Run the code

```
woo@secr2043:~$ bash woo.sh
cp: missing destination file operand after '../cars/*.*.'
Try 'cp --help' for more information.
cp: missing destination file operand after '../fruits/*.*.'
Try 'cp --help' for more information.
```

Then draw the tree

```
woo@secr2043:~$ tree
.
├── WOOCHENGSHUAN
├── bye.txt
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── greeting.txt
├── ifiles.sh
├── ifles.sh
└── woo.sh

6 directories, 18 files
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 ifiles.sh
#!/bin/bash

#prompt the user to enter a file extension
read -p "Enter the file extension (without the dot): " ext

#Find and display all files with the given extension
echo "Files with ${ext} extension: "
find . -type f -name "*${ext}"

#count the number of files with the given extension
count=$(find .type f -name "*${ext}" |wc -l)
echo "Number of files with ${ext} extension: $count"

woo@secr2043:~$ chmod u+x ifiles.sh
woo@secr2043:~$ ./ifiles.sh
Enter the file extension (without the dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

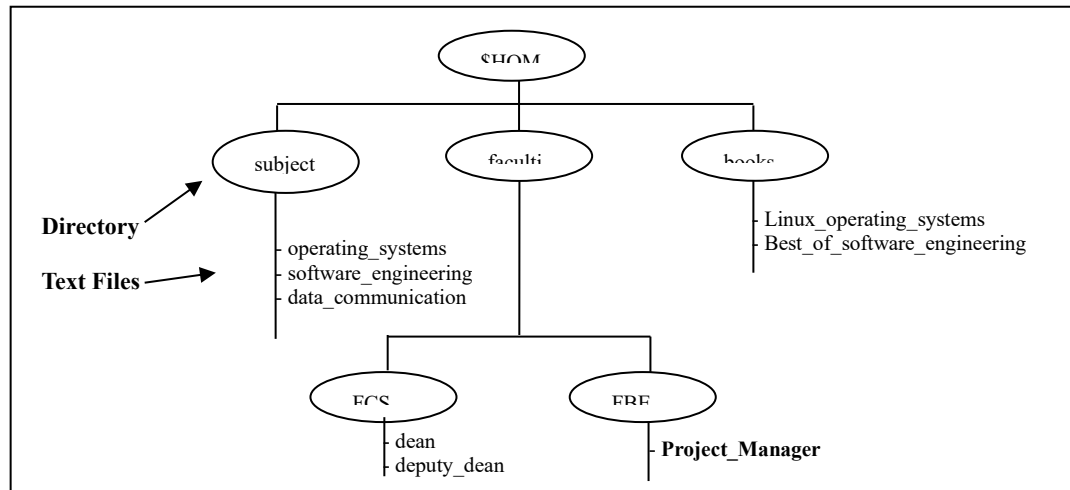


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text file contains its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2                                woo2a.sh
#!/bin/bash

#Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

#Create and write to text files in the subject directory
echo "operating system" >$HOME/subjects/operating_systems
echo "software engineering" >$HOME/subjects/software_engineering
echo "data communication" >$HOME/subjects/data_communication

#Create and write to text files in the faculties/FCS directory
echo "dean" >$HOME/faculties/FCS/dean
echo "deputy dean" >$HOME/faculties/FCS/deputy_dean

#Create and write to text files in the faculties/FBE directory
echo "Project Manager" >$HOME/faculties/FBE/Project_Manager

#Create and write to text files in the books directory
echo "Linux operating system" >$HOME/books/Linux_operating_systems
echo "Best of software engineering" >$HOME/books/Best_of_software_engineering

woo@secr2043:~$ chmod u+x woo2a.sh
woo@secr2043:~$ ./woo2a.sh
woo@secr2043:~$ tree $HOME
/home/woo
├── WOOCHENGSHUAN
├── books
│   ├── Best_of_software_engineering
│   └── Linux_operating_systems
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── faculties
│   ├── FBE
│   │   └── Project_Manager
│   └── FCS
│       ├── dean
│       └── deputy_dean
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── ifiles.sh
├── subjects
│   ├── data_communication
│   ├── operating_systems
│   └── software_engineering
├── woo.sh
└── woo2a.sh
11 directories, 24 files

```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--



```

woo@secr2043:~$ ls -ld books
drwxrwxr-x 2 woo woo 4096 Jun 22 09:22 books
woo@secr2043:~$ ls -ld subjects
drwxrwxr-x 2 woo woo 4096 Jun 22 09:22 subjects
woo@secr2043:~$ ls -ld faculties/FCS
drwxrwxr-x 2 woo woo 4096 Jun 22 09:22 faculties/FCS
woo@secr2043:~$ ls -ld books/Best_of_software_engineering
-rw-rw-r-- 1 woo woo 29 Jun 22 09:22 books/Best_of_software_engineering
woo@secr2043:~$ ls -ld faculties/FBE/Project_Manager
-rw-rw-r-- 1 woo woo 16 Jun 22 09:22 faculties/FBE/Project_Manager

```

- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

GNU nano 7.2                                woo2c.sh
#!/bin/bash

#Set permission for subjects directory
chmod 744 $HOME/subjects

#Set permission for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

#Set permission for FCS directory
chmod 607 $HOME/faculties/FCS

#Set permission for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager

woo@secr2043:~$ chmod +x woo2c.sh
woo@secr2043:~$ ./woo2c.sh
woo@secr2043:~$ ls -ld ~/subjects
drwxr--r-- 2 woo woo 4096 Jun 22 09:22 /home/woo/subjects
woo@secr2043:~$ ls -ld ~/faculties/FCS
drw---rwx 2 woo woo 4096 Jun 22 09:22 /home/woo/faculties/FCS
woo@secr2043:~$ ls -ld ~/books/Best_of_software_engineering
-r-x-w--- 1 woo woo 29 Jun 22 09:22 /home/woo/books/Best_of_software_engineering
woo@secr2043:~$ ls -ld ~/faculties/FBE/Project_Manager
-----wx-x 1 woo woo 16 Jun 22 09:22 /home/woo/faculties/FBE/Project_Manager

```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c). [2 marks]

Directory/File	Access Control
subjects	drwxr-r--
Best_of_software_engineering	drw----rwx
FCS	-r-x-w----
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****